

TABLE I
CONTINUED

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Controls</i>							
Bin-state FE	Y	Y	Y	Y	Y	Y	
Bin-period FE	Y	Y	Y	Y	Y	Y	
Bin-state linear trends		Y	Y		Y	Y	
Bin-state quadratic trends			Y			Y	
Bin-division-period FE				Y	Y	Y	
State FE							Y
Year FE							Y

Notes. The table reports the effects of a minimum wage increase based on the event study analysis (see equation (1)) exploiting 138 state-level minimum wage changes between 1979 and 2016. The table reports five-year averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs at and up to \$5 above it, employment, and wages. Column (1) shows the benchmark specification while columns (2)–(6) explore robustness to bin-state time trends and bin-division-period fixed effects. Column (7) reports the simpler methodology estimates where we calculate changes in affected wage and employment by using state-by-quarter data, where the outcomes are the number of jobs or the total wage bill under \$15 per hour. Regressions are weighted by state-quarter aggregated population. Robust standard errors in parentheses are clustered by state. Significance levels are *0.10, **0.05, ***0.01.

The first two rows report the change in number of missing jobs below the new minimum wage (Δb), and excess jobs above the new minimum wage (Δa) relative to the pretreatment total employment. The third row, the percentage change in average wages in the affected bins, ($\% \Delta W$), is calculated using equation (2). The fourth row, percentage change in employment in the affected bins, is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{b - 1}$). The fifth row, employment elasticity with respect to the minimum wage, is calculated as $\frac{\Delta a + \Delta b}{\% \Delta W}$ whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{b - 1}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

between 5.7% and 6.9% with common wage-bin-period effects (columns (1)–(3)), and between 4.3% and 5.0% with division-specific wage-bin-period effects (columns (4)–(6)). In contrast, the proportionate change in employment for affected workers is never statistically significant, and is numerically smaller than the wage change, ranging between -1.9% and 3.6% across the eight specifications. For the most part, the employment estimates are small or positive; the only exception is column (5) with state-specific linear trends and bin-division-specific period effects. The employment elasticity with respect to the wage is -0.449 (std. err. 0.574). However, adding quadratic trends to the former specification (column (6)) substantially reduces the magnitude of the employment elasticity with respect to the wage to -0.003 (std. err. 0.455).

Finally, column (7) provides employment and wage estimates using a state-by-period panel, where we regress either the per capita wage bill or employment under an absolute wage threshold ($\bar{W}$), and then estimate the change in affected wage and employment using the same formulae as our baseline.²¹ The estimates and standard errors for affected employment (0.025, std. err. 0.029) and wage (0.063, std. err. 0.011) are virtually identical to column (1), clarifying that use of wage bins or choices around those have no impact on our key estimates. At the same time, unlike our baseline specification, this simpler method using an absolute wage threshold cannot provide separate estimates for excess and missing jobs.

[Online Appendix Table A.4](#) shows that our results are robust to focusing only on the events occurring in the states that do not allow tip credits; dropping occupations that allows tipping; using full-time equivalent job counts; restricting the sample to hourly workers; additionally using federal-level minimum wage changes for identification; using the raw CPS data instead of the QCEW benchmarked CPS; without using population weights; or focusing on the post-1992 period. We also show robustness to alternative event window lengths ([Online Appendix Table A.6](#)), and alternative values of the upper endpoint of the wage window, $\bar{W}$ ([Online Appendix Table A.5](#)).

21. The threshold is $W = 15$, which is at least \$4 above the new minimum wage in all of our events but one.

III.A. *Heterogenous Responses to the Minimum Wage*

We can use our approach focused on low-wage jobs to estimate the effect of the minimum wage on specific subgroups.

1. *By Demographic Groups.* We assess the presence of labor-labor substitution at the bottom of the wage distribution by examining employment responses across various demographic groups.²² In Table II we report estimates for workers without a high school degree, those with high school or less schooling, women, black or Hispanic individuals, and teens using our baseline specification (see equation (1)).

As expected, restricting the sample by education and age produces a larger bite. For example, for those without a high school degree, the missing jobs estimate, Δb , is -6.5% while for those with high school or less schooling it is -3.2% . These estimates for the missing jobs are, respectively, 261% and 78% larger than the baseline estimate for the overall population (-1.8% , from Table I, column (1)). Nevertheless, the large variation in the missing jobs across various demographic groups is matched closely by excess jobs above the new minimum wage.²³ In all cases, except for the black or Hispanic group, the excess jobs are larger than the missing jobs, indicating a positive albeit statistically insignificant employment effect. For black or Hispanic individuals, the difference between excess and missing jobs is negligible. As a result, the employment elasticities with respect to own wage range between -0.086 and 0.570 for the first five demographic groups of the table. In all cases but one, the elasticities are statistically indistinguishable from 0. The sole exception is those without a high school degree, for whom the employment elasticity with respect to the wage is 0.475 (std. err. 0.268) which is marginally significant at the ten percent level. The minimum wage elasticity for teens is 0.125 , which is more positive than some of the estimates in the

22. Existing evidence on labor-labor substitution has typically focused on specific groups like teens (Giuliano 2013), individual case studies (Fairris and Bujanda 2008), or specific segments like online labor platforms (Horton 2018).

23. In Online Appendix Figure A.8 we also show that the close match between excess jobs and missing jobs holds also if we fully partition the workforce into 23 age-education cells.

TABLE II
IMPACT OF MINIMUM WAGES ON EMPLOYMENT AND WAGES BY DEMOGRAPHIC GROUPS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Missing jobs below new MW (Δb)	-0.065*** (0.010)	-0.032*** (0.007)	-0.114*** (0.010)	-0.023*** (0.005)	-0.028*** (0.008)	-0.094*** (0.010)	-0.020*** (0.005)	-0.004*** (0.001)
Excess jobs above new MW (Δa)	0.075*** (0.011)	0.038*** (0.006)	0.127*** (0.020)	0.026*** (0.004)	0.028*** (0.006)	0.100*** (0.012)	0.021*** (0.003)	0.004*** (0.001)
% Δ affected wages	0.080*** (0.014)	0.076*** (0.014)	0.083*** (0.018)	0.072*** (0.011)	0.044*** (0.012)	0.073*** (0.011)	0.051*** (0.013)	0.060* (0.032)
% Δ affected employment	0.038 (0.024)	0.043 (0.030)	0.030 (0.032)	0.025 (0.027)	-0.004 (0.044)	0.015 (0.018)	0.015 (0.048)	0.011 (0.055)
Employment elasticity w.r.t. MW	0.097 (0.061)	0.061 (0.042)	0.125 (0.134)	0.025 (0.027)	-0.005 (0.058)	0.052 (0.062)	0.016 (0.049)	0.003 (0.014)
Emp. elasticity w.r.t. affected wage	0.475* (0.268)	0.570 (0.386)	0.356 (0.317)	0.343 (0.362)	-0.086 (1.005)	0.206 (0.233)	0.304 (0.904)	0.184 (0.841)
Jobs below new MW ($\bar{b}_{-1}$)	0.264	0.145	0.432	0.102	0.133	0.358	0.104	0.027
% Δ MW	0.103	0.101	0.102	0.101	0.100	0.103	0.103	0.103